

Education

- 2024 – Now **Hong Kong Polytechnic University**, Ph.D. in Computing, GPA: 4.1/4.3.
Advisor: Prof. **Wenqi Fan** and Prof. **Qing Li**
- 2021 – 2024 **Sichuan University**, M.E. in Computer Science and Technology, GPA: 3.76/4.
Advisor: Prof. **Min Zhu**, Lab: Vision Computing Lab
- 2017 – 2021 **Sichuan University**, B.E. in Computer Science and Technology, GPA: 3.77/4.
Under the Wu Yuzhang Honors Program

Research Interests

AI for Science, Protein Language Models, Large Language Models, Diffusion Models

Publications & Preprints (First Author)

- 2026 **Yi Zhou**, Qipeng Wang, Yunqing Liu, Jun Xia, Qing Li, Wenqi Fan. "Unlocking Multimodal Protein Language Models at Inference Time". *Conference on Empirical Methods in Natural Language Processing (EMNLP Main)*, 2026. (CCF-B)
- Systematically studied the inference design space of multimodal protein language models by exploring three bottom-up strategies (vanilla sampling, classifier-free guidance, and reward-guided beam search), achieving substantial parameter-free performance gains and revealing base model behaviors.
- Yi Zhou**, Haohao Qu, Yunqing Liu, Shanru Lin, Le Song, Wenqi Fan. "HD-Prot: A Protein Language Model for Joint Sequence-Structure Modeling with Continuous Structure Tokens". *International Conference on Knowledge Discovery and Data Mining (KDD)*, 2026. (CCF-A)
- Proposed HD-Prot, a *Hybrid Diffusion Protein* language model that integrates continuous structure tokens into a discrete protein language model via a unified diffusion process, achieving competitive and efficient protein sequence-structure joint modeling.
- 2025 **Yi Zhou**, Wenqi Fan, Shijie Wang, Yuyao Yan, Hui Liu, Qian Zhao, Le Song, Qing Li. "Computational Protein Science in the Era of Large Language Models (LLMs)". *ACM Computing Surveys (CSUR)*, Under Review.
- Authored a comprehensive review on protein language models, systematically summarizing their foundational sequence-structure-function knowledge modeling and discussing their adaptations, applications and future directions in protein structure prediction, function prediction, and protein design.
- 2024 **Yi Zhou**, Xian Guan, Meixuan Wu, Chengzhou Ouyang, Min Zhu. "Timely-MDA: A Benchmark for Generalizable MiRNA-Disease Association Prediction". *International Conference on Bioinformatics and Biomedicine (BIBM)*, 2024. (CCF-B)
- 2022 **Yi Zhou**, Xinyi Wang, Lin Yao, Min Zhu. "LDAformer: Predicting LncRNA-Disease Associations based on Topological Feature Extraction and Transformer Encoder". *Briefings in Bioinformatics (BIB)*, 2022. (JCR-Q1, IF: 13.994, CCF-B)

Publications & Preprints (Collaboration)

- 2026 Yunqing Liu, **Yi Zhou**, Wenqi Fan. "Enhancing Molecular Property Predictions by Learning from Bond Modelling and Interactions". *International Conference on Learning Representations (ICLR)*, 2026. (CCF-A)

Yunqing Liu, **Yi Zhou**, Jun Xia, Wenqi Fan, Qing Li. "Geometric Flow Matching for Conformation Generation via Manifold Decomposition". *IEEE Transactions on Knowledge and Data Engineering (TKDE)*, 2026, Under Review.

Tianzhong Lan, Weili Jiang, Yisong Liu, **Yi Zhou**, Junqi Bai, Si Yong Yeo, Xulei Yang, Min Zhu. "Wavelet-Driven Cross-Domain Consistency for Mixed-Supervised 3D Tumor Segmentation". *European Conference on Computer Vision (ECCV)*, 2026. (**CCF-B**)

2025 Lin Gan, Xinyi Wang, **Yi Zhou**, Min Zhu. "Protein-binding RNA Prediction Based on Integrated Sequence-Structure-Function Pre-training". *IEEE Transactions on Computational Biology and Bioinformatics (TCBB)*, 2025. (**JCR-Q1**)

Meixuan Wu, Yizhou Yang, **Yi Zhou**, Kai Wang, Junqi Bai, Min Zhu. "PLHGMDA: Pre-trained Language Model and Heterogeneous Graph Neural Network for MiRNA-Disease Association Prediction". *International Conference on Intelligent Computing (ICIC)*, 2025. (**CCF-C, Oral**)

2024 Wenwen Gao, Shangsong Liu, **Yi Zhou**, Fengjie Wang, Feng Zhou, Min Zhu. "GBDT4CTRVis: Visual Analytics of Gradient Boosting Decision Tree for Advertisement Click-Through Rate Prediction". *Journal of Visualization (JoV)*, 2024.

Jiamin Zhu, Meixuan Wu, **Yi Zhou**, Nan Cao, Haotian Zhu, Min Zhu. "Dowsing: A Task-Driven Approach for Multiple-View Visualizations Dynamic Recommendation". *Journal of Visualization (JoV)*, 2024.

Tutorials

2026 Yujuan Ding, Liangbo Ning, Ziran Liang, Chun-Hin Chan, **Yi Zhou**, Haohao Qu, Wenqi Fan. "Towards Autonomous Web Interaction: Empowering WebAgents by Large Foundation Models". *Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD)*, 2026. (**CCF-C**)

Teaching & Services

2023 - 2024 **Teaching Assistant**, Sichuan University.

- Data Visualization (311301030) × 2

2024 - Now **Teaching Assistant**, Hong Kong Polytechnic University.

- Introduction to Artificial Intelligence and Data Analytic (COMP1004) × 3

- Computational Thinking and Problem Solving (COMP1010)

Reviewer.

- Conference: KDD AI for Sciences Track - 2026/2027

- Journal: Expert Systems With Applications (ESWA)

Honors & Awards

2024 PolyU Research Postgraduate Scholarship (PRPgS)

Outstanding Graduate of Sichuan University

Outstanding Undergraduate Teaching Assistant at Sichuan University

2023 Outstanding Graduate Student of Sichuan University

First-class of Excellent Graduate Scholarship by Sichuan University

Tencent Scholarship of Sichuan University

2021 Certificate of Honor from Wu Yuzhang Honors College

Skills

Programming Python, PyTorch, SQL, Linux commands, Git, LaTeX

Languages Mandarin, English (IELTS: 7.0)